

ACTIVITY 19 — LEARNER WORKSHEET**Kanban Board with WIP Limits and Lead/Cycle Time**

TYPE Analysis	DURATION 45 minutes	MAPS TO LO5	TOPIC 5 - Monitor and Control the Project
-------------------------	-------------------------------	-----------------------	---

What you are doing

Contoso Learning Pte Ltd is a training provider with 14,000 active learners. Its course registration portal is 9 years old. Registration takes a median of 11 minutes and 34% of attempts are abandoned. Learner communications consume 24 staff-hours per week across 3 administrators. 62% of learner traffic is mobile, against a portal that is not responsive. Legacy consent screens do not meet current PDPA requirements. The project modernises course registration and learner communications.

What you will produce: kanban board, flow metrics, bottleneck analysis — submitted to artifacts/.

What you need before you start

- Lab 10 product backlog and story points
- Lab 08 definition of done
- Lab 15 RACI

Your steps

1. Design the workflow columns. A three-column To Do / Doing / Done board hides everything interesting. Contoso's real workflow has distinct stages, and the waiting states between them are where the delay actually lives.
2. Set WIP limits. Open Kanban Board and create the columns above with these WIP limits.
3. Load the real card data. These are the 18 stories completed by the Contoso team during sprints 8 and 9, with the dates each card entered the system, started development, and was accepted. All dates are working days measured from the start of sprint 8 as day 1.
4. Compute the flow metrics
5. Apply Little's Law
6. Find the bottleneck. A bottleneck reveals itself as a queue that grows in front of one stage. Here is the average number of cards in each column, sampled daily across the 20 days.
7. Read the cumulative flow diagram. The Kanban tool will render a cumulative flow diagram. Learn to read it, because a CFD is a strong candidate for a graphic-based exam item.
8. Answer the exam-style scenarios

Your working space

Complete every field. Keep this worksheet — it is your reference for the rest of the course.

Step 1 — Design the workflow columns

Step 2 — Set WIP limits

Step 3 — Load the real card data

Step 4 — Compute the flow metrics

Step 5 — Apply Little's Law

Step 6 — Find the bottleneck

Step 7 — Read the cumulative flow diagram

Step 8 — Answer the exam-style scenarios

Deliverable checklist

- 19-kanban-board.md - the nine-column workflow with entry and exit policies per column and a justified WIP limit for each, with reasoning tied to actual team composition.
- A screenshot or export of your Kanban tool board with the 18 cards loaded.
- 19-flow-metrics.md - mean and median lead time, mean and median cycle time, throughput per day and per sprint, and the flow efficiency percentage, all with working shown.
- The Little's Law verification against observed WIP on a sample day, plus the three predictive calculations (WIP raised to 12, WIP cut to 4, and the 22-card release forecast).
- 19-bottleneck-analysis.md - the column utilisation table, the identified bottleneck with at least three pieces of supporting evidence, and the five Theory of Constraints steps applied to it.
- A written statement of why raising the WIP limit at the bottleneck is the wrong response.

- Your reading of the cumulative flow diagram, naming what the widening band indicates.
- Written answers to the three scenarios.